

UART TRANSMITTER

RTL-to-GDSII Implementation

A Complete Digital VLSI Physical Design Project

Project	UART-TX-RTL-to-GDSII
Design Type	Sequential / Clocked RTL Design
Technology	Sky130
Implementation Flow	OpenLane v1.0.2
Top Module	uart_tx
Physical Design	Synthesis → Floorplan → Placement → CTS/Clock Analysis → Routing → Signoff
Final Deliverable	GDSII layout with signoff reports

Final Project Report

Prepared from the completed UART project implementation and its recorded design/signoff results.

1. Abstract

This project implements a UART transmitter as a sequential digital design and carries it through a complete RTL-to-GDSII physical design flow using the Sky130 technology and OpenLane v1.0.2. The project was developed as a clocked design so that, in addition to RTL functionality and physical implementation, clock-aware timing analysis and multi-corner static timing analysis could be demonstrated.

The completed flow includes RTL development, simulation, waveform verification, logic synthesis, floorplanning, I/O placement, power planning, placement, clock-related implementation/analysis, routing, static timing analysis, physical signoff, and final GDSII generation. The final implementation produced GDSII, KLayout-compatible GDS, LEF, Liberty, SDF, SPICE and MAG outputs together with DRC, LVS, XOR, antenna and timing reports.

2. Project Objectives

- Design and verify a practical sequential UART transmitter at RTL.
- Demonstrate a clocked VLSI design rather than only a combinational block.
- Run the design through an end-to-end RTL-to-GDSII flow using OpenLane.
- Generate a physical layout using the Sky130 standard-cell technology.
- Perform timing analysis including detailed multi-corner STA.
- Complete physical signoff checks and generate a final GDSII database.
- Organize implementation results, reports and evidence as a GitHub portfolio project.

3. Design Overview

The design is a UART transmitter (UART TX). UART is an asynchronous serial communication protocol in which a byte is transmitted serially over a TX line using a defined bit period. The transmitter is controlled by a clock and internally sequences the transmitted frame.

3.1 Main Design Characteristics

Item	Project implementation
Design	UART transmitter
Top-level module	uart_tx
Design class	Sequential / clocked
Clock	Used by the transmitter state/bit sequencing
Serial output	TX output
Reset/control	Included as part of the RTL control structure
Physical implementation	Standard-cell based Sky130 flow

3.2 Functional Operation

- The transmitter receives a byte/control request at RTL.
- A clock drives the sequential transmission process.
- The UART frame is serialized onto the TX output.
- The transmitter returns to its idle state after the frame is completed.
- RTL simulation was used to verify the expected serial waveform before physical design.

3.3 RTL-to-GDSII Flow

The implementation follows the standard digital physical-design sequence: RTL → synthesis → floorplan → I/O placement → power planning → placement → clock-related implementation/analysis → routing → extraction/STA → signoff → GDSII.

4. RTL Verification and Simulation

The UART RTL was verified before physical implementation. A dedicated testbench was used to exercise the transmitter and generate a waveform dump. The project contains simulation evidence and a GTKWave waveform image in the screenshots directory.

Verification Evidence

Evidence	Project file/evidence
UART RTL	uart_tx RTL source
Testbench	uart_tx testbench
Simulation	UART simulation result
Waveform	GTKWave waveform evidence
Physical layout	Final GDS/KLayout layout evidence

The simulation stage establishes functional confidence at RTL before the design is passed to synthesis and physical implementation. The waveform screenshot is retained as project evidence together with the RTL and simulation screenshots.

5. Physical Design Implementation

5.1 Synthesis

The UART RTL was synthesized into a gate-level standard-cell representation using the OpenLane flow and Sky130 standard-cell libraries. The synthesis stage generated area/timing-related reports and the gate-level representation required for subsequent physical design.

5.2 Floorplanning and I/O Placement

The design was assigned a physical core/die region and the I/O interface was placed for the subsequent power, placement and routing stages.

5.3 Power Planning and Placement

Power distribution structures were generated and the synthesized standard cells were placed within the floorplan. The placement results were preserved in the project repository as implementation evidence.

5.4 Clock-Related Implementation

Unlike the earlier combinational ALU project, this UART design is clocked. Therefore clock-aware analysis and CTS-related implementation are applicable to the design.

5.5 Routing

The placed design was routed to create physical interconnects between standard cells and I/O pins. The completed run generated the routed DEF and ODB databases as well as final extracted/signoff artifacts.

6. Physical Design Results

The recorded UART implementation results are summarized below.

Metric	Recorded result
Standard-cell count	182
Total cell area	1923.094400 μm^2
Total power	5.00e-04 W
Internal power	3.57e-04 W
Switching power	1.43e-04 W
Leakage power	1.64e-09 W
TNS	0.00
WNS	0.00
Worst setup slack	6.63
Worst hold slack	0.31

The reported zero TNS and zero WNS indicate that the recorded final timing summary did not contain negative aggregate slack. The setup and hold slack values above are the recorded final signoff summary values used for this project report.

7. Static Timing Analysis (STA)

Static timing analysis was completed for the clocked UART design. The project preserves detailed STA material rather than only a single timing number. The repository contains separate maximum/minimum/nominal STA reports and a combined all-corners report.

7.1 STA Report Organization

Report	Purpose
rcx_max_sta.*	RCX-based maximum-corner timing analysis
rcx_min_sta.*	RCX-based minimum-corner timing analysis
rcx_nom_sta.*	RCX-based nominal-corner timing analysis
STA_report_all_corners.rpt	Combined detailed multi-corner STA report
31-rcx_sta.*	Final signoff STA report set

7.2 Timing Summary

Timing item	Final recorded result
TNS	0.00
WNS	0.00
Worst setup slack	6.63
Worst hold slack	0.31
STA coverage	Detailed max/min/nominal and combined all-corner reports retained

The detailed all-corner STA report is retained as generated project evidence. This report does not replace or alter the STA output; it documents that the timing analysis was performed and that the detailed timing artifacts are part of the final project.

8. Physical Signoff

The final implementation passed the recorded physical signoff checks. The signoff evidence is stored in the project results directory.

Signoff check	Recorded result
DRC	0 violations
LVS	0 errors
XOR / layout comparison	0 differences
Antenna	No listed violations in final antenna report
Final GDSII	Generated successfully
KLayout GDS	Generated successfully

8.1 Final Generated Artifacts

- `uart_tx.gds` — final GDSII database
- `uart_tx.klayout.gds` — KLayout-compatible GDS
- `uart_tx.lef` — layout exchange format
- `uart_tx.lib` — timing/power library representation
- `uart_tx.sdf` — extracted timing information
- `uart_tx.spice` — extracted SPICE representation
- `uart_tx.mag` — Magic layout representation
- `uart_tx.rpt` — final implementation report

9. Project Repository Organization

The UART project is maintained as a separate GitHub repository so that it can be presented independently from the previously completed 4-bit ALU project.

Directory / item	Purpose
src / RTL	UART transmitter source
simulation	Testbench and simulation material
results/floorplan	Floorplan-related reports/artifacts
results/placement	Placement DEF/ODB and reports
results/routing	Routing DEF/ODB and routing reports
results/sta	Detailed STA and all-corner timing reports
results/signoff	DRC/LVS/XOR/antenna and final signoff reports
results/final	Final GDSII and extracted deliverables
screenshots	RTL, simulation, waveform and layout evidence
docs	Final project documentation

The repository was pushed to GitHub and the recorded working tree was clean after the physical-design result organization. The project therefore has both reproducible implementation artifacts and presentation-oriented evidence.

10. Technical Learning and Outcomes

- Implemented a sequential UART transmitter at RTL.
- Understood the transition from behavioral RTL to synthesized standard-cell logic.
- Worked through the major OpenLane physical-design stages.
- Handled floorplanning, placement, routing and clock-related implementation for a sequential design.
- Generated and organized multi-corner STA reports.
- Performed physical signoff checks including DRC, LVS and XOR comparison.
- Generated industry-standard physical-design deliverables such as GDSII, LEF, LIB, SDF and SPICE.
- Organized the complete project as a GitHub portfolio repository.

11. Conclusion

The UART transmitter project successfully demonstrates an end-to-end digital VLSI implementation flow from RTL through physical signoff and GDSII generation. The project is stronger than a purely combinational example because it includes a clocked sequential design and corresponding timing analysis. The completed implementation includes functional verification, physical design databases, detailed multi-corner STA material and signoff evidence.

The final result is a complete portfolio-ready RTL-to-GDSII project demonstrating practical exposure to RTL design, simulation, synthesis, physical implementation, timing analysis and physical verification using Sky130 and OpenLane.

End of Report